

Are Prompt Optimizers Blind? Cross-Modal Visual Feedback for Automatic Prompt Optimization

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Abstract

Automatic prompt optimization (APO) has been widely adopted to adapt vision-language models (VLMs) to downstream tasks without weight updates, yielding promising results. However, on multimodal tasks the effectiveness of APO is fundamentally bottlenecked by a *blind feedback channel*: the optimizer reads the question, the prediction, and the gold answer, but never the input image on which the model failed, and therefore cannot diagnose visually grounded errors. As a remedy, we introduce **Cross-Modal Visual Feedback (CMVF)**. CMVF incorporates (1) a failure-conditioned visual diagnosis stage, in which a stronger optimizer VLM inspects each failed image *without access to predictions or labels*, and (2) an error-aware aggregation stage that compresses these observations into reusable, task-level visual blind-spot patterns that drive the prompt rewrite. Crucially, the image is consumed *only* during optimization; the deployed artifact is an ordinary text prompt that runs at the same inference cost as any text-only baseline. Extensive results across 12 VQA datasets and 4 target VLMs demonstrate that CMVF consistently ranks first, improving over the strongest baseline on every target by **+2.4** points on average (up to **+6.5** on individual benchmarks), while the optimizer self-organizes into expert-style visual checklists that transfer across models without re-optimization. Our code and optimized prompts will be released.

1 Introduction

Vision-language models (VLMs) are the default substrate for visual question answering, medical imaging, chart understanding, and document interpretation (Masry et al. 2022; Mathew, Karatzas, and Jawahar 2021), yet their accuracy is notoriously sensitive to the wording of the textual prompt—on our benchmarks, mean accuracy on Qwen3.5-4B varies from 70.3% to 75.0% across competitive prompts alone. Automatic prompt optimization (APO)—spanning *evolutionary search* (Fernando et al. 2024; Guo et al. 2024), *textual-gradient methods* (Pryzant et al. 2023; Yuksekogonul et al. 2025), *response-trajectory tracking* (Zhang et al. 2025), and *declarative program compilation* (Khatab et al. 2024)—has become an effective, weight-free way to adapt these models, yielding promising results on text-centric tasks.

However, when applied to multimodal tasks these optimizers all share the same structural gap: their feedback channel

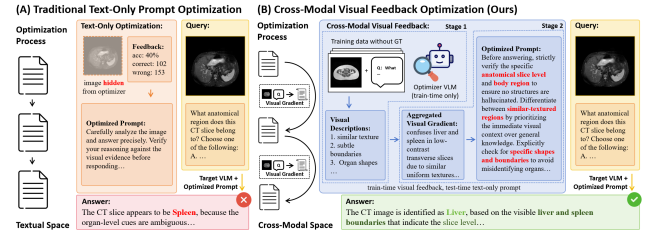


Figure 1: Motivation for CMVF. Existing text-only APO sees scores and error triples but misses the visual causes of failures. CMVF uses failed images only during optimization to aggregate visual blind spots into a reusable text prompt, with no extra visual-feedback calls at test time.

is *blind*. The optimizer sees the question, the prediction, and the gold answer, but never the image on which the target failed (Figure 1)—like a teacher grading an exam without ever seeing the figures. It can register *that* an error occurred but not *why*: a confused anatomical slice level, an overlooked axis label, or a global scene prior that overrides local evidence. We argue that the bottleneck of multimodal APO is not the optimization machinery but the *evidence* available to it.

As a remedy, we introduce **Cross-Modal Visual Feedback (CMVF)**, built on a simple principle: *optimization feedback must have access to the modality in which failures originate*. CMVF lets a stronger optimizer VLM inspect the failed images during optimization—first describing question-relevant visual evidence without seeing the prediction or ground-truth answer, then aggregating these descriptions with the error triples into reusable visual blind spots that drive the prompt rewrite. Unlike multimodal prompt optimization (MPO) (Choi et al. 2026), which jointly optimizes a (text, image) prompt pair and keeps the image in the loop at deployment, CMVF pays for the visual signal *only once*, at optimization time, and compiles it into a purely textual prompt; deployment therefore adds no VLM calls and reuses the same serving interface as any text-only prompt.

Our contributions are three-fold:

- 1. Problem.** We identify and characterize the *visual-feedback blindness* shared by all text-only APO methods on multimodal tasks, and show the bottleneck is *evidence*,

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not optimization machinery.

2. **Method.** We propose CMVF, which incorporates (1) failure-conditioned, label-free visual diagnosis and (2) error-aware aggregation into reusable visual blind-spot patterns, realizing a *train-time-multimodal, deploy-time-text-only* design at zero added inference cost.
3. **Results & analysis.** Across 12 datasets and 4 target VLMs, CMVF ranks first everywhere, beating the strongest baseline by +2.4 pp on average (up to +6.5 pp); token-budget-matched ablations attribute the gain to visual perception, the optimizer self-organizes into *expert-style visual checklists*, and these strategies transfer across models without re-optimization.

2 Cross-Modal Visual Feedback

The preceding discussion raises two questions that define the design of CMVF. *First*, how can a prompt optimizer obtain visual diagnostic information without leaking the ground-truth answer into the visual analysis? *Second*, how can this visual information improve the deployed prompt without adding any inference-time VLM calls? CMVF answers these two questions with a two-stage visual-gradient channel (Section 2.3) and a fusion-update loop that preserves a standard textual-prompt inference interface (Section 2.4).

2.1 Problem Setup

Given a target VLM $\mathcal{M}_T$ and training data $\mathcal{D} = \{(x_i, q_i, y_i)\}_{i=1}^N$ (x_i image, q_i question, y_i ground-truth answer), we seek a text prompt $p^* \in \mathcal{P}$ that maximizes task accuracy:

$$p^* = \arg \max_{p \in \mathcal{P}} \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\mathcal{M}_T(p, x_i, q_i) = y_i]. \quad (1)$$

Following TextGrad (Yuksekgonul et al. 2025), we optimize two prompts jointly: p_{yn} for yes/no questions and p_{oe} for open-ended questions, both with a $\{\text{fs}\}$ placeholder for few-shot examples. At step $t \in \{1, \dots, T\}$, let $\hat{y}_i := \mathcal{M}_T(p_t, x_i, q_i)$ be the prediction under the current prompt $p_t = (p_{yn}^{(t)}, p_{oe}^{(t)})$; we form the correct set $\mathcal{C}_t$ and the wrong set $\mathcal{W}_t = \{(x_i, q_i, \hat{y}_i, y_i) : \hat{y}_i \neq y_i\}$. A validation set $\mathcal{V}$ selects the best prompt across steps, and if a rewrite drops the $\{\text{fs}\}$ placeholder we restore the validation-best prompt.

2.2 The Visual-Feedback Bottleneck

Standard APO constructs feedback from accuracy and textual error triples, $g_t^{\text{ext}} = (\text{acc}(p_t), \mathcal{E}_t^{\text{sub}})$ with $\mathcal{E}_t^{\text{sub}} \subseteq \{(q_i, \hat{y}_i, y_i)\}$. This exposes *which* questions failed but omits the visual evidence that caused them: a medical VQA model may confuse adjacent anatomical slice levels, a chart model may miss a small axis label, a real-world model may rely on a global scene prior. The textual triple sees the symptom; the cause stays hidden. CMVF closes this loop with the two-stage visual channel below.

2.3 Visual Diagnostic Channel

CMVF adds a visual-gradient channel to the feedback loop. During optimization only, a stronger optimizer VLM $\mathcal{M}_O$

inspects failed images and summarizes recurring visual blind spots. Figure 2 gives the overview.

Stage 1: question-aware visual description. For every wrong example, $\mathcal{M}_O$ receives only the image and question and produces a question-aware visual description:

$$v_i = \mathcal{M}_O(x_i, q_i; \pi_{\text{vis}}), \quad \forall (x_i, q_i, \hat{y}_i, y_i) \in \mathcal{W}_t, \quad (2)$$

where π_{vis} is a fixed Stage-1 meta-prompt asking for question-relevant visual evidence (supplementary material). Crucially, the prediction $\hat{y}_i$ and ground truth y_i are *withheld* at this stage. This prevents answer-conditioned rationalization—the optimizer cannot manufacture a post-hoc explanation for a label it has not yet seen—and forces it to record perception-level evidence rather than rationalize a known label.

Stage 2: error-aware aggregation. Individual descriptions are noisy and idiosyncratic. We therefore pair each v_i with its error triple and ask $\mathcal{M}_O$ to summarize recurring blind spots:

$$g_t^{\text{vis}} = \text{Agg}(\{(q_i, \hat{y}_i, y_i, v_i) : (x_i, q_i, \hat{y}_i, y_i) \in \mathcal{W}_t\}). \quad (3)$$

The aggregator receives at most 30 wrong cases per step with field-level truncation. The two-stage decomposition separates visual perception from supervised error attribution: Stage 1 records *what is visible*; Stage 2 identifies *which visual patterns repeatedly correlate with mistakes*, compressing them into a compact set of reusable diagnostic instructions (e.g., “check small text,” “verify anatomical slice level,” “avoid global scene priors”). This compression is what makes broad inspection at Stage 1 affordable: many descriptions in, one prompt-sized signal out.

2.4 Fusion, Prompt Update, and Inference

Fused feedback. We fuse accuracy statistics and visual blind spots into a unified feedback string:

$$g_t^{\text{CM}} = \text{CONCAT}(\text{acc}(p_t), |\mathcal{C}_t|, |\mathcal{W}_t|, g_t^{\text{vis}}, s), \quad (4)$$

where s instructs the optimizer to rewrite the prompt so the target model explicitly checks the identified visual features before answering. Because g_t^{vis} is constructed from both visual descriptions and textual error triples, CMVF *augments* rather than replaces the existing textual error signal.

Update and inference. The prompt is updated by $p_{t+1} = \text{UPDATE}(p_t, g_t^{\text{CM}})$. The default UPDATE is a TextGrad-style rewrite (Yuksekgonul et al. 2025); CMVF* swaps in a REVOLVE-style alternative (Zhang et al. 2025) on the same visual-gradient pipeline, isolating whether gains come from the visual channel rather than a particular rewriter. At deployment, the target VLM receives only the image, question, and optimized text prompt (p_{yn}, p_{oe}) —no additional visual analysis at inference (Algorithm 1).

2.5 Modeling Intuition

The role of visual feedback can be understood through a simple error-cause decomposition. Let $\alpha_V \in [0, 1]$ denote the fraction of currently wrong examples whose failure has a visual component, and let $I_V \in [0, 1]$ denote how

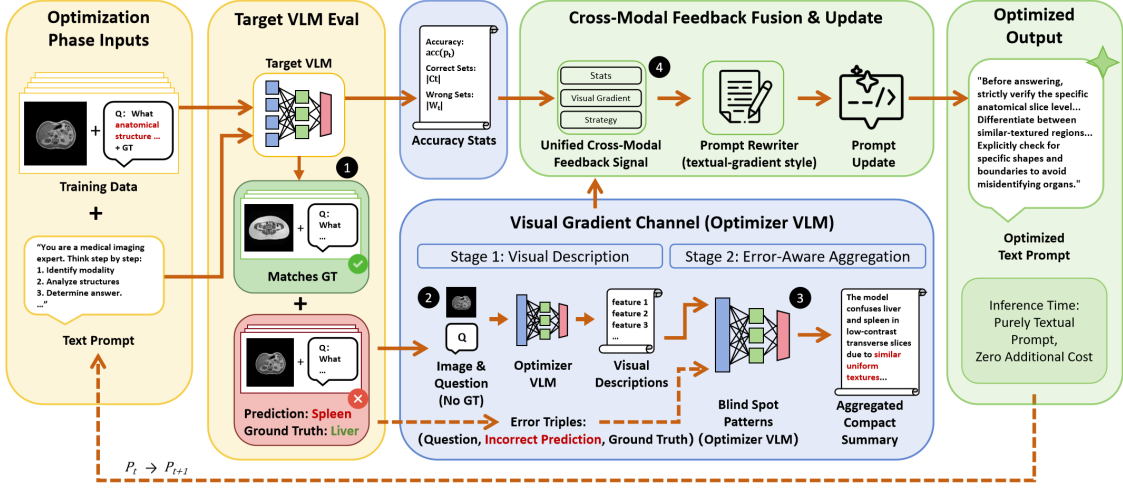


Figure 2: CMVF optimization pipeline. The target VLM is evaluated on training examples; the optimizer VLM inspects failed images, aggregates visual blind spots, and rewrites the text prompt. Inference uses the optimized text prompt with no extra visual-feedback calls.

Algorithm 1 CMVF Prompt Optimization

Require: Target VLM $\mathcal{M}_T$, optimizer VLM $\mathcal{M}_O$, training set $\mathcal{D}$, validation set $\mathcal{V}$, steps T

- 1: Initialize prompts (p_{yn}, p_{oe}) and validation-best prompts
- 2: **for** $t = 1$ to T **do**
- 3: Evaluate $\mathcal{M}_T$ on $\mathcal{D}$ to obtain $\mathcal{C}_t$ and $\mathcal{W}_t$
- 4: Evaluate on $\mathcal{V}$ and update the validation-best prompts if improved
- 5: **for each** $(x_i, q_i, \hat{y}_i, y_i) \in \mathcal{W}_t$ **do**
- 6: $v_i \leftarrow \mathcal{M}_O(x_i, q_i; \pi_{vis})$ {image + question only}
- 7: **end for**
- 8: $g_t^{vis} \leftarrow \text{AGG}(\{(q_i, \hat{y}_i, y_i, v_i) : (x_i, q_i, \hat{y}_i, y_i) \in \mathcal{W}_t\})$
- 9: $g_t^{CM} \leftarrow \text{CONCAT}(\text{acc}(p_t), |\mathcal{C}_t|, |\mathcal{W}_t|, g_t^{vis}, s)$
- 10: $(p_{yn}, p_{oe}) \leftarrow \text{UPDATE}((p_{yn}, p_{oe}), g_t^{CM})$
- 11: Restore validation-best prompts if $\{fs\}$ is lost
- 12: **end for**
- 13: **return** validation-best prompts

informative the optimizer’s visual analysis is about these causes. Feedback without visual inspection carries only indirect visual hints; denote its residual visual signal by δ_T . With error rate $L(p) := 1 - \text{acc}(p)$ and per-step reduction $\Delta L := L(p_t) - L(p_{t+1})$, under an additive view of textual and visual error sources the expected advantage of cross-modal feedback scales as

$$\mathbb{E}[\Delta L_{CM} - \Delta L_{text}] \gtrsim \alpha_V I_V - \delta_T. \quad (5)$$

This intuition (not a convergence guarantee) predicts CMVF should help most when α_V is large (medical VQA, TextVQA OCR, RealWorldQA) and least when $\alpha_V \rightarrow 0$; both match our empirical pattern in Section 3.2. This also clarifies the role of Stage-2 aggregation. A single visual description may be noisy or instance-specific, so its information about the underlying visual cause is limited. By aggregating many failed

cases, CMVF suppresses instance-specific noise, surfaces recurring visual patterns, and converts per-image observations into a compact prompt-level signal. In this sense, aggregation increases the *effective* I_V , which is consistent with the ablation results in Section 3.5.

3 Experiments

In this section, we run experiments to address five questions:

- **Q1:** Does CMVF beat APO baselines that do not incorporate visual feedback, across multiple target VLMs? (Section 3.2)
- **Q2:** Does the optimizer learn anything *interpretable* about the failure modes? (Section 3.3)
- **Q3:** Do the optimized strategies *transfer* across target models without re-optimization? (Section 3.4)
- **Q4:** Where do the gains come from—image-aware perception, or extra feedback tokens? (Section 3.5)
- **Q5:** Does CMVF generalize and pay an acceptable training cost with unchanged inference? (Section 3.6)

Stage-2 variants (CM-Mem, Diff), dataset details, prompt templates, and case studies are in the Appendix.

3.1 Setup

Models and benchmarks. We use Qwen3.5-9B as the optimizer VLM $\mathcal{M}_O$ and evaluate four target VLMs: Qwen3.5-4B, Qwen2.5-VL-7B (Wang et al. 2024), LLaVA-1.6-Mistral-7B (Li et al. 2023), and Phi-3.5-vision (Abdin et al. 2024). The suite covers 12 datasets: medical VQA datasets SLAKE CT/MRI/X-Ray, VQA-RAD, and PathVQA (Liu et al. 2021; Lau et al. 2018; He et al. 2020); reasoning and chart/OCR datasets ScienceQA, ChartQA, TextVQA, and MMMU (Lu et al. 2022; Masry et al. 2022; Singh et al. 2019; Yue et al. 2024); and RealWorldQA, AI2D, and DocVQA (xAI 2024; Kembhavi et al. 2016; Mathew, Karatzas, and Jawahar 2021). We focus the main compari-

son on eight benchmarks and report the remaining four (ScienceQA, AI2D, DocVQA, MMMU) separately in Table 2.

Baselines and protocol. We compare against ProTeGi (Pryzant et al. 2023), OPRO (Yang et al. 2024), DSPy (Khattab et al. 2024), TextGrad (TG) (Yuksekgonul et al. 2025), and REVOLVE (RE) (Zhang et al. 2025). Each benchmark uses 150 training samples, 150 validation samples, and the full test set. All methods share data splits, initialization prompts, the same optimizer VLM, and 20 optimization steps. Results are averaged over 3 runs. Evaluation follows each benchmark’s official scoring protocol (exact match after lowercasing and punctuation removal, plus a small set of domain synonyms, e.g., “x-ray”/“xray”).

3.2 A1: CMVF Ranks First on All Four Target VLMs

Headline result. Table 1 restores the full comparison across all five baselines without visual feedback, CMVF, and CMVF* for all four target VLMs. **CMVF ranks first on mean accuracy for every target**, with the largest absolute gain on LLaVA-1.6-Mistral-7B (+3.1 pp) and a comparable +3.0 pp on Qwen3.5-4B.

Where the wins concentrate. The per-dataset pattern is consistent with the visual-feedback hypothesis: gains cluster on visually demanding benchmarks. On Qwen3.5-4B, CMVF improves RealWorldQA by **+6.5 pp**, TextVQA by **+2.4 pp**, and PathVQA by **+1.5 pp** over the strongest baseline without visual feedback. On LLaVA, ChartQA jumps by **+6.0 pp** and PathVQA by **+3.3 pp**. On the four additional benchmarks, CMVF again ranks first, by +1.12 to +3.88 pp over the strongest baseline (Table 2). This matches the modeling intuition in Equation (5): the gain scales with α_V , the fraction of visually-grounded errors, which is largest precisely on perception-heavy tasks.

The gains are statistically significant. Exact McNemar tests against the strongest baseline, on paired per-example predictions, confirm the improvements: $p = 7.9 \times 10^{-14}$ on LLaVA/TextVQA, $p = 0.021$ on LLaVA/ChartQA, and $p = 0.015$ on Qwen2.5-VL/TextVQA—significant precisely on OCR, chart-grounding, and fine-grained visual tasks, where visual evidence is essential.

It is not the rewriter. CMVF* uses a REVOLVE-style rewriter on top of the same visual-gradient pipeline. Relative to its counterpart without visual feedback, adding the visual-gradient channel improves a REVOLVE-style update on most targets (e.g., 70.3→71.9 on Qwen3.5-4B, 54.7→58.5 on LLaVA, and 53.7→55.8 on Phi). This shows that the visual channel is not tied to a TextGrad-style rewriter, while the remaining gap to CMVF also makes clear that the rewriter choice still matters.

3.3 A2: The Optimizer Self-Discovers Expert-Style Visual Inspection Checklists

Beyond the headline accuracy gains, we observe a striking qualitative effect: **the optimizer VLM is not given dataset-specific visual failure rules, yet it converges on instruc-**

tions that look like a domain expert’s checklist. This emergent self-diagnostic behavior is, in our view, the most interesting observation of this paper.

Prompt contrast and source audit. Table 3 lists representative deployed prompts on Qwen3.5-4B. Baselines without visual feedback converge on *generic, vision-agnostic* hedges (“read step by step,” “be precise”), while CMVF produces *task-specific visual checklists*. Crucially, this is not an artifact of hand-written rules: the Stage-1 prompt is deliberately broad (“key diagnostic finding” for medical, “key visual details” for general) and never mentions chart axes, small-text OCR, slice-level checks, or local-object priors—yet the final prompts recover exactly these task-level routines from failed-image evidence.

What CMVF rewrites the prompt to say. Feedback without visual inspection typically yields generic advice (“verify your reasoning against the visual evidence,” “be careful with details”). CMVF instead discovers *task-specific* routines across the 8 benchmarks: (1) SLAKE-CT/MRI prompts ask the model to verify anatomical slice level and separate similar-density regions; (2) VQA-RAD/PathVQA prompts locate the focal abnormality before classification; (3) ChartQA prompts read axes, units, and legends before comparing values; (4) TextVQA prompts scan for small or non-salient text; and (5) RealWorldQA prompts check occlusion and avoid global scene priors for local-object questions. These instructions are not pre-specified in the seed prompt or Stage-1 description prompt; they arise from Stage-2 aggregation over Stage-1 perception, showing that a frozen optimizer VLM can self-organize reusable visual checklists from failed-image evidence alone.

Why this matters. The phenomenon suggests that the bottleneck of multimodal APO is not optimization machinery but *evidence*: once the optimizer can see failed images, recurring visual failure modes become extractable in language and can be carried by a static inference-time text prompt.

3.4 A3: Optimized Strategies Transfer Across Target Models

If CMVF merely memorized target-specific quirks, its prompts would not survive being moved to a different model. We test this directly: the CMVF prompt optimized on each of Qwen2.5-VL-7B, LLaVA-1.6-7B, and Phi-3.5-vision is deployed *unchanged* on Qwen3.5-4B, with *no* re-optimization (Table 4). Every transferred prompt outperforms a TextGrad prompt optimized *directly* on Qwen3.5-4B (70.3), by +1.2 to +2.8 pp on the 8-benchmark mean, and the advantage holds per-dataset rather than on a few benchmarks—on ChartQA a transferred prompt even exceeds the in-domain one. This indicates that Stage-2 aggregation surfaces reusable, *task-level* visual strategies rather than spurious correlations bound to a single model, directly addressing the concern that access to ground-truth labels could induce dataset- or target-specific overfitting. The in-domain prompt optimized on 4B remains best overall (75.0), as expected.

Model	Method	Medical VQA ↑					General VQA ↑			Mean ↑
		CT	MRI	X-Ray	RAD	Path	Chart	Text	RWQA	
 Qwen3.5-4B	ProTeGi	75.3 $\pm$ 2.8	78.1 $\pm$ 0.9	81.7 $\pm$ 1.4	59.4 $\pm$ 3.1	45.4 $\pm$ 0.5	78.0 $\pm$ 3.4	65.2 $\pm$ 2.3	80.4 $\pm$ 1.9	70.4
	OPRO	73.2 $\pm$ 3.4	67.7 $\pm$ 6.1	81.8 $\pm$ 1.0	55.2 $\pm$ 4.7	44.6 $\pm$ 1.9	25.1 $\pm$ 28.4	69.1 $\pm$ 0.6	81.7 $\pm$ 3.3	62.3
	DSPy	<u>77.6</u> $\pm$ 1.1	<u>77.7</u> $\pm$ 1.4	<u>80.7</u> $\pm$ 1.3	<u>62.1</u> $\pm$ 0.8	<u>47.5</u> $\pm$ 2.8	81.6 $\pm$ 1.9	<u>66.3</u> $\pm$ 2.2	<u>82.6</u> $\pm$ 0.9	<u>72.0</u>
	TG	75.9 $\pm$ 0.3	75.3 $\pm$ 1.3	80.7 $\pm$ 1.6	57.9 $\pm$ 3.1	45.4 $\pm$ 2.0	78.6 $\pm$ 2.7	63.8 $\pm$ 1.4	<u>84.8</u> $\pm$ 0.7	70.3
	RE	75.5 $\pm$ 0.6	<u>79.0</u> $\pm$ 0.1	<u>83.8</u> $\pm$ 0.5	59.2 $\pm$ 2.0	44.3 $\pm$ 1.1	75.1 $\pm$ 2.7	64.3 $\pm$ 1.6	<u>80.9</u> $\pm$ 2.2	70.3
	CMVF*	76.8 $\pm$ 0.4	<u>77.9</u> $\pm$ 2.2	<u>84.1</u> $\pm$ 0.2	61.3 $\pm$ 1.4	46.9 $\pm$ 0.8	78.8 $\pm$ 2.0	65.7 $\pm$ 0.8	<u>83.3</u> $\pm$ 1.8	71.9
	CMVF	77.9 $\pm$ 1.3	80.7 $\pm$ 3.1	84.5 $\pm$ 1.2	63.4 $\pm$ 1.0	49.0 $\pm$ 2.8	81.4 $\pm$ 2.0	71.5 $\pm$ 2.7	91.3 $\pm$ 3.8	75.0 ↑3.0
 Qwen2.5-VL-7B	ProTeGi	64.0 $\pm$ 1.3	59.4 $\pm$ 4.9	<u>73.5</u> $\pm$ 1.1	47.5 $\pm$ 2.3	37.9 $\pm$ 0.9	56.3 $\pm$ 0.6	59.8 $\pm$ 0.6	58.3 $\pm$ 11.0	57.1
	OPRO	60.1 $\pm$ 6.4	61.4 $\pm$ 8.3	69.4 $\pm$ 7.5	42.6 $\pm$ 0.1	29.5 $\pm$ 4.4	54.2 $\pm$ 3.8	60.0 $\pm$ 0.2	50.4 $\pm$ 8.0	53.4
	DSPy	60.8 $\pm$ 1.1	61.0 $\pm$ 2.8	72.1 $\pm$ 1.3	47.0 $\pm$ 0.6	32.7 $\pm$ 1.3	59.4 $\pm$ 2.1	55.5 $\pm$ 2.6	60.0 $\pm$ 8.6	56.1
	TG	<u>64.5</u> $\pm$ 2.0	<u>69.2</u> $\pm$ 1.3	<u>73.5</u> $\pm$ 1.6	48.8 $\pm$ 1.6	36.8 $\pm$ 1.9	54.4 $\pm$ 1.9	59.4 $\pm$ 1.8	<u>61.9</u> $\pm$ 2.7	<u>58.6</u>
	RE	63.5 $\pm$ 0.2	<u>69.4</u> $\pm$ 0.6	72.1 $\pm$ 0.2	48.3 $\pm$ 1.0	<u>38.6</u> $\pm$ 0.9	<u>57.5</u> $\pm$ 0.3	57.1 $\pm$ 3.7	58.7 $\pm$ 2.1	58.2
	CMVF*	63.8 $\pm$ 0.9	68.7 $\pm$ 3.2	73.2 $\pm$ 2.2	49.9 $\pm$ 1.4	35.4 $\pm$ 2.7	55.7 $\pm$ 1.2	59.3 $\pm$ 1.5	<u>61.9</u> $\pm$ 0.7	58.5
	CMVF	65.0 $\pm$ 1.1	69.4 $\pm$ 2.3	75.0 $\pm$ 3.0	51.0 $\pm$ 1.8	40.2 $\pm$ 2.2	56.1 $\pm$ 2.9	62.3 $\pm$ 1.7	62.2 $\pm$ 3.2	60.1 ↑1.6
 LLaVA-1.6-7B	ProTeGi	58.4 $\pm$ 0.9	72.7 $\pm$ 0.1	<u>76.8</u> $\pm$ 0.4	45.6 $\pm$ 0.6	38.6 $\pm$ 0.2	25.5 $\pm$ 0.9	45.3 $\pm$ 1.8	81.6 $\pm$ 1.2	55.6
	OPRO	58.3 $\pm$ 1.6	65.6 $\pm$ 0.3	71.9 $\pm$ 1.2	41.1 $\pm$ 0.8	36.3 $\pm$ 0.2	32.3 $\pm$ 3.7	49.5 $\pm$ 2.4	80.1 $\pm$ 0.5	55.4
	DSPy	58.5 $\pm$ 0.6	66.9 $\pm$ 1.1	74.5 $\pm$ 0.9	44.3 $\pm$ 1.7	37.8 $\pm$ 0.7	31.4 $\pm$ 5.2	49.4 $\pm$ 3.0	80.3 $\pm$ 1.3	54.4
	TG	58.9 $\pm$ 2.4	70.1 $\pm$ 2.6	75.7 $\pm$ 0.2	43.8 $\pm$ 1.1	34.9 $\pm$ 2.0	30.9 $\pm$ 2.1	45.9 $\pm$ 3.4	76.1 $\pm$ 6.8	54.5
	RE	56.4 $\pm$ 1.2	69.6 $\pm$ 2.8	76.1 $\pm$ 0.5	45.1 $\pm$ 0.3	37.6 $\pm$ 1.4	29.1 $\pm$ 1.1	43.6 $\pm$ 1.8	80.4 $\pm$ 0.6	54.7
	CMVF*	<u>60.3</u> $\pm$ 0.3	<u>70.9</u> $\pm$ 1.0	<u>75.8</u> $\pm$ 0.9	<u>48.0</u> $\pm$ 2.5	<u>40.6</u> $\pm$ 0.2	<u>37.7</u> $\pm$ 3.7	<u>53.5</u> $\pm$ 1.5	81.0 $\pm$ 0.2	<u>58.5</u>
	CMVF	59.6 $\pm$ 0.6	71.1 $\pm$ 0.4	77.7 $\pm$ 0.6	48.2 $\pm$ 0.1	41.9 $\pm$ 3.3	38.3 $\pm$ 4.4	51.3 $\pm$ 1.9	81.3 $\pm$ 0.4	58.7 ↑3.1
 Phi-3.5-vision	ProTeGi	58.2 $\pm$ 2.1	<u>72.0</u> $\pm$ 4.9	76.0 $\pm$ 0.6	45.5 $\pm$ 2.4	32.5 $\pm$ 1.6	50.9 $\pm$ 1.8	47.4 $\pm$ 1.7	45.8 $\pm$ 2.0	53.5
	OPRO	<u>59.2</u> $\pm$ 1.0	63.8 $\pm$ 2.5	69.5 $\pm$ 2.2	40.4 $\pm$ 2.9	31.9 $\pm$ 0.8	45.0 $\pm$ 6.4	49.3 $\pm$ 2.0	48.1 $\pm$ 3.9	50.9
	DSPy	57.4 $\pm$ 1.7	64.0 $\pm$ 1.4	74.1 $\pm$ 2.1	42.6 $\pm$ 1.3	34.4 $\pm$ 1.0	51.1 $\pm$ 1.8	51.2 $\pm$ 1.5	55.7 $\pm$ 6.8	53.8
	TG	57.7 $\pm$ 2.1	68.5 $\pm$ 2.6	75.0 $\pm$ 0.7	43.0 $\pm$ 2.3	34.1 $\pm$ 1.1	51.2 $\pm$ 1.6	49.5 $\pm$ 2.1	45.7 $\pm$ 5.0	53.1
	RE	56.4 $\pm$ 0.8	68.6 $\pm$ 2.1	75.0 $\pm$ 1.2	43.3 $\pm$ 0.6	37.1 $\pm$ 0.8	48.6 $\pm$ 1.1	51.8 $\pm$ 0.7	48.6 $\pm$ 3.7	53.7
	CMVF*	60.1 $\pm$ 1.1	71.9 $\pm$ 0.9	<u>77.3</u> $\pm$ 1.6	<u>46.4</u> $\pm$ 3.0	38.5 $\pm$ 1.0	54.4 $\pm$ 2.6	52.3 $\pm$ 1.4	45.2 $\pm$ 1.2	<u>55.8</u>
	CMVF	58.9 $\pm$ 1.9	72.4 $\pm$ 1.5	77.1 $\pm$ 2.3	46.5 $\pm$ 0.5	38.0 $\pm$ 1.5	53.4 $\pm$ 1.1	51.4 $\pm$ 1.9	49.8 $\pm$ 1.4	55.9 ↑2.1

Table 1: Accuracy on 8 benchmarks. Rows group target VLMs; per model the **CMVF** (our method) row is shaded green and the runner-up by mean gray. **Bold/underline** mark best/second best; gray subscripts show std over 3 runs.

Method	AI2D	DocVQA	MMMU	SciQA	Mean
ProTeGi	78.46	68.31	<u>47.62</u>	81.42	68.95
OPRO	76.83	64.28	45.89	<u>85.30</u>	68.08
DSPy	74.21	66.74	43.47	78.95	65.84
TG	72.68	62.53	41.16	76.63	63.25
RE	<u>80.91</u>	<u>70.86</u>	<u>47.62</u>	83.07	<u>70.62</u>
CMVF	84.62	74.74	48.74	87.60	73.93

Table 2: Accuracy (%) on the four additional benchmarks (Qwen3.5-4B). The **CMVF** row is highlighted; the runner-up by mean (RE) is shaded gray. **Bold**: best; underline: best baseline (ProTeGi and RE tie on MMMU). CMVF ranks first on all four and on their mean.

3.5 A4: Image-Aware Perception Drives the Gains, Not Extra Tokens

Tokens are cheap; pixels are decisive. A2 attributed the gains to image-aware evidence; Table 5 now rules out longer feedback as the cause under a token-budget-matched protocol on Qwen3.5-4B. **CM-Blind** adds the same number of feedback tokens as CMVF but *without image access*; it reaches only 71.9 (+1.6 pp over TG). **CM-Caption** feeds the optimizer generic, question-agnostic captions and reaches 72.2. **CM-Vis** keeps the visual channel but drops the fused error attribution—concatenating raw per-image descriptions

into the rewrite prompt, closely matching the IPO-style (Du, Sun, and Snoek 2024) recipe adapted from CLIP class-template prompts to VQA instruction prompts—reaching 72.8, while the full **CMVF** adds Stage-2 aggregation over *all* wrong samples and reaches **75.0**. The implication is direct: longer prompts alone account for at most +1.6 pp over TG, while *question-aware visual perception with aggregation* accounts for the remaining +3.1 pp. CMVF’s *deployed* prompt averages ~205 BPE tokens (excluding { f s }), comparable to TG/RE (185–192) and 14–26% shorter than DSPy/OPRO/ProTeGi (239–276); the gain cannot be attributed to a longer inference-time prompt.

The ordering generalizes across targets. The same monotone channel progression—CM-Blind < CM-Caption < CM-Vis < CMVF—holds on *every* (target, dataset) pair for two additional target VLMs (Table 5), so the effect is not specific to Qwen3.5-4B. The margin widens on the cross-family target LLaVA-1.6-7B (CMVF +7.8 over CM-Vis on Real-WorldQA and +7.5 on TextVQA), consistent with a weaker base model leaving more visually-groundable headroom for the visual channel to recover.

Two alternative Stage-2 aggregation modes further isolate *why* unified aggregation is preferred: **CM-Mem** (72.1, −2.9 vs. CMVF), which carries a cross-step memory pool and is misled by stale signals from earlier weaker prompts, and **Diff** (72.2, −2.8), which fragments the visual signal by partitioning errors before aggregation. Both underperform

Method	Deployed prompt (visual-specific clauses highlighted)	Acc
■ TextVQA (scene text, OCR)		
OPRO	<i>Imagine your mind is a vast, still pond where ripples of distraction fade away until only the single drop of truth answering the question can be seen; gently surface to inspect this one isolated ripple. . .</i>	69.1
TG	<i>Analyze the image to extract only visual information, explicitly ignoring any text, labels, or metadata that cannot be verified by visual inspection alone.</i>	63.8
CMVF	Meticulously scan the entire image to verify specific counts, read small text, and confirm fine details rather than relying on generic associations or prominent but unrelated features.	71.5 ↑2.4
■ ChartQA (chart reasoning, axes/legends)		
TG	<i>Examine the chart and answer the question. Be precise and double-check your interpretation before giving the final answer.</i>	78.6
RE	<i>Analyze the chart step by step, revise any inconsistent interpretation, and return the final value or category.</i>	75.1
CMVF	Read axis labels, units, and the legend mapping before reasoning about magnitudes; identify the requested data point and verify its exact value before answering.	81.4 ↑2.8
■ RealWorldQA (real-world VQA, scene reasoning)		
DSPy	<i>You are a visual question answering assistant. Inspect the image and answer the question accurately.</i>	82.6
TG	<i>Look at the image carefully and answer the question. Make sure your reasoning is grounded in what is actually visible.</i>	<u>84.8</u>
CMVF	Scan for small, low-contrast, occluded, or background objects; verify orientation, signs, and object states; avoid global scene priors when the question is about a local object.	91.3 ↑6.5

Table 3: Representative deployed prompts on Qwen3.5-4B. For each benchmark we show two baselines (without visual feedback, gray italics) and CMVF (green). **Bold** marks the decisive clause: baselines stay generic or, on TextVQA, even instruct the model to ignore text, whereas CMVF names the exact visual check the task demands. Acc is test accuracy.

Prompt source → Qwen3.5-4B	CT	MRI	X-Ray	RAD	Path	Chart	Text	RWQA	Mean
TextGrad (direct on 4B)	75.9	75.3	80.7	57.9	45.4	78.6	63.8	84.8	70.3
CMVF ← Qwen2.5-VL-7B	76.0	76.8	82.5	57.4	44.8	83.2	66.1	84.8	71.5
CMVF ← LLaVA-1.6-7B	76.4	78.7	83.4	59.4	48.2	80.5	65.5	84.3	72.1
CMVF ← Phi-3.5-vision	76.0	79.0	83.0	58.3	47.6	84.3	69.7	87.0	73.1
CMVF (direct on 4B)	77.9	80.7	84.5	63.4	49.0	81.4	71.5	91.3	75.0

Table 4: Cross-target transfer: a CMVF prompt optimized on one model and deployed on Qwen3.5-4B *without* re-optimization still beats TextGrad optimized directly on 4B. Every transferred prompt exceeds the direct TextGrad baseline; on ChartQA transfer even beats the in-domain prompt. Bold: best per column.

Method	Qwen3.5-4B	Qwen2.5-VL-7B		LLaVA-1.6-7B	
	Mean	RWQA	Text	RWQA	Text
CM-Blind	71.9	56.1	56.1	69.1	39.5
CM-Caption	72.2	58.3	58.2	71.3	41.5
CM-Vis	72.8	60.5	60.4	73.5	43.8
CMVF	75.0	62.2	62.3	81.3	51.3

Table 5: Feedback-channel ablation across three target VLMs (methods defined in text; token-budget matched). *Qwen3.5-4B* is the 8-benchmark mean; the other two targets are per-dataset (official protocol). Every column rises monotonically—CMVF (green) best, CM-Vis (gray) runner-up.

unified CMVF (full design-space sweep in the appendix), confirming that the gain comes from question-aware perception of *all* wrong samples with a single unified aggregation step, not from any individual component.

3.6 A5: CMVF Generalizes Better at ~25% Training-Time Overhead and No Additional Inference Cost

Late-stage improvement. Figure 3 shows that CMVF *continues* improving after baselines without visual feedback flatten on these settings. Once feedback without visual inspection has corrected the linguistically obvious failures, the remaining errors are more likely to be visually grounded, so an image-grounded gradient can keep moving.

Generalization gap. Figure 4(a) shows that CMVF has the *smallest* average validation-to-test gap among all six methods, and Figure 4(b) confirms this advantage holds per-dataset rather than coming from a few easy benchmarks. We attribute this to aggregation: instead of overfitting prompts to isolated validation errors, CMVF rewrites them around *recurring* visual patterns—a form of implicit regularization.

Cost in practice. The visual-gradient pipeline adds a one-shot training-time overhead but *no additional* inference-time cost. End-to-end optimization on Qwen3.5-4B takes **2.6 h** for CMVF versus 2.1 h for TextGrad and 2.4 h for REVOLVE—a ~20–25% wall-clock increase that buys +3.0 pp mean accu-

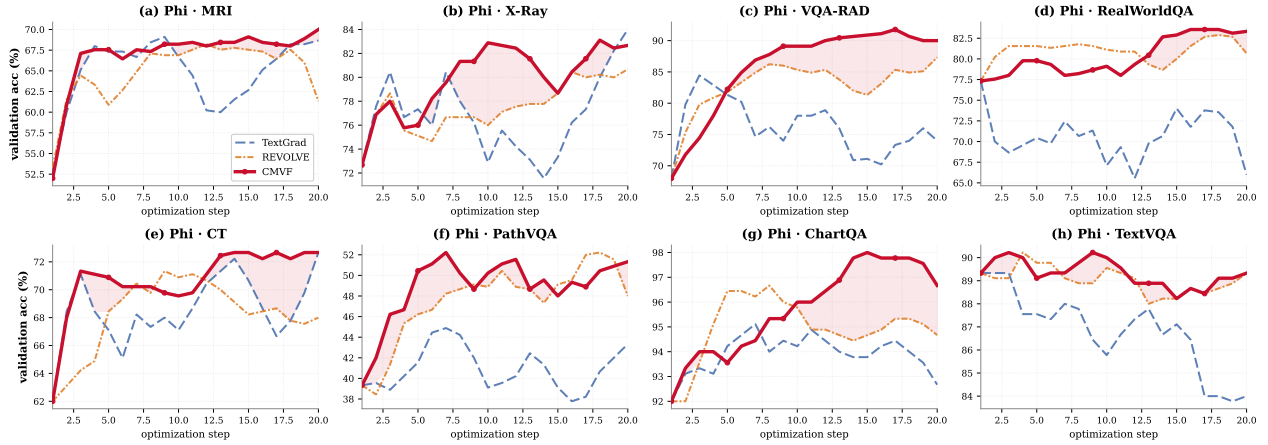


Figure 3: Validation accuracy over 20 optimization steps on eight Phi-3.5-vision benchmarks (top: MRI, X-Ray, VQA-RAD, RealWorldQA; bottom: CT, PathVQA, ChartQA, TextVQA). CMVF (red) stays above both text-only baselines (TextGrad, REVOLVE) and keeps improving after they plateau; the shaded band marks CMVF’s margin over the stronger baseline.

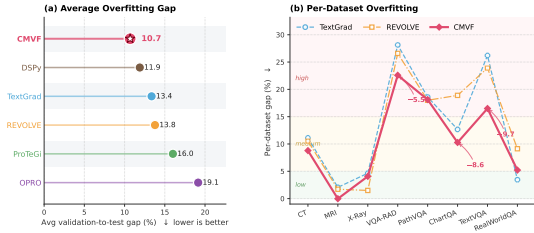


Figure 4: Generalization gap (validation-best accuracy minus test accuracy; lower is better) on 8 benchmarks. CMVF has the smallest average gap and reduces high-gap cases such as ChartQA and VQA-RAD.

racy. Compared to ProTeGi (5.9 h), CMVF is more than $2\times$ faster while delivering substantially higher accuracy. Crucially, the final deployed artifact is a standard text prompt, so inference uses the *same serving interface and call count* as a vanilla deployment.

4 Related Work

Automatic prompt optimization. Early APO methods search over prompt candidates or use LLMs as meta-optimizers (Yang et al. 2024; Zhou et al. 2023), including reinforcement-learning, black-box, and evolutionary variants (Deng et al. 2022; Chen et al. 2024; Fernando et al. 2024; Guo et al. 2024). DSPy compiles declarative LLM programs into optimized prompt pipelines (Khatab et al. 2024), gradient-inspired methods such as ProTeGi and TextGrad use natural-language critiques as textual gradients (Pryzant et al. 2023; Yuksekgonul et al. 2025), and REVOLVE tracks response evolution across iterations (Zhang et al. 2025). As prompts are highly sensitive to surface form (Sclar et al. 2024), recent work also explores symbolic and structure-factorized prompt spaces (Ou et al. 2025; Liu et al. 2026). These methods keep the feedback channel textual even when the downstream task is multimodal.

Prompt learning for vision-language models. A parallel line adapts VLMs by learning *continuous* prompts: CoOp and CoCoOp tune soft context vectors for CLIP-style classifiers (Zhou et al. 2022b,a), VPT prepends learnable visual tokens (Jia et al. 2022), and MaPL couples prompts across the vision and language branches (Khatab et al. 2023). These require gradient access to the backbone and deploy a non-textual prompt, whereas CMVF optimizes a *discrete, human-readable* instruction and treats the target VLM as a black box at both optimization and deployment time.

Multimodal prompting, visual feedback, and self-refinement. The closest prior work is IPO (Du, Sun, and Snoek 2024), which lets an LMM describe training images and rewrites a CLIP class-template prompt for zero-shot classification. CMVF shares its recipe of *answer-agnostic visual descriptions + LMM optimizer + textual-prompt deployment*, but inspects only the wrong set $\mathcal{W}_t$, adds a Stage-2 error-aware aggregation into reusable blind-spot patterns (removing it costs 2.2 pp; Table 5), and targets generative VQA prompts on four VLMs (4B–7B) rather than a fixed CLIP backbone. Concurrently, *multimodal prompt optimization* (MPO) (Choi et al. 2026) keeps the deployed prompt multimodal at a per-query visual cost, whereas CMVF deploys a standard textual prompt. Unlike self-refinement methods that critique outputs per instance at inference time (Madaan et al. 2023; Shinn et al. 2023), CMVF aggregates cross-example visual evidence *once*, at optimization time.

5 Conclusion

We introduce CMVF, a two-stage method that lets the optimizer VLM inspect the images the target model failed on and compiles that evidence into an ordinary text prompt. CMVF ranks first on all four target VLMs, beating the strongest baseline by +2.4 pp on average (up to +6.5 pp) at no added inference cost, and the optimizer self-organizes into transferable expert-style visual checklists. The bottleneck of multimodal APO is *visual evidence*, not optimization machinery.

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